

Continuum Mechanics

class # 07

9/8/2026

Curl Operator

Definition (2.12): To any vector $\vec{v}: \mathbb{E}^3 \rightarrow \mathcal{V}$ we associate another vector field

$$\nabla \times \vec{v}: \mathbb{E}^3 \rightarrow \mathcal{V}$$

defined by

$$(\nabla \times \vec{v}) \times \vec{a} = (\nabla \vec{v} - \nabla \vec{v}^T) \vec{a}$$

for all constant vectors $\vec{a} \in \mathcal{V}$. We call $\nabla \times \vec{v}$ the curl of $\vec{v}$.

Remark: At a point $\vec{x}$, if $\vec{v}$ is vector velocity field $\nabla \times \vec{v}(\vec{x})$ describes the rate and direction of the rotation at $\vec{x}$.

Result (2.13): let $\mathcal{F} = \{\vec{e}_i\}$ be an arbitrary frame and let $\vec{v}(\vec{x}) = v_i(\vec{x}) \vec{e}_i$. Then,

$$(\nabla \times \vec{v})(\vec{x}) = \epsilon_{ijk} v_{i,k}(\vec{x}) \vec{e}_j,$$

where (x_1, x_2, x_3) are the coordinates of $\vec{x}$ in $\{\vec{e}_i\}$.
Equivalently,

$$\begin{aligned} \nabla \times \vec{v} = & \left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3} \right) \vec{e}_1 + \left(\frac{\partial v_1}{\partial x_3} - \frac{\partial v_3}{\partial x_1} \right) \vec{e}_2 \\ & + \left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \right) \vec{e}_3 \end{aligned}$$

Proof: Recall Result (1.3). If W is skew symmetric, there is a unique axial vector $\vec{w}$ such that

$$\vec{w} \times \vec{a} = W\vec{a}, \quad \forall a \in V.$$

Since $\nabla\vec{v} - \nabla\vec{v}^T$ is skew symmetric, there is a unique vector $\nabla \times \vec{v}$ such that

$$(\nabla \times \vec{v}) \times a = (\nabla\vec{v} - \nabla\vec{v}^T)a \quad \forall a \in V.$$

Thus, the curl is well defined. Set $\vec{w} = \nabla \times \vec{v}$ and $W = \nabla\vec{v} - \nabla\vec{v}^T$. By Result (1.3) we have

$$w_j = \frac{1}{2} \epsilon_{ijk} W_{ik} = \frac{1}{2} \epsilon_{ijk} (v_{i,k} - v_{k,i})$$

Recall that

$$\epsilon_{ijk} = -\epsilon_{kji}$$

$$w_j = \frac{1}{2} \epsilon_{ijk} v_{i,k} - \frac{1}{2} \epsilon_{ijk} v_{k,i}$$

$$= \frac{1}{2} \epsilon_{ijk} v_{i,k} - \frac{1}{2} \epsilon_{kji} v_{i,k} \quad \left(\begin{matrix} \text{curved arrow} \\ i \rightarrow k \end{matrix} \right)$$

$$= \frac{1}{2} \epsilon_{ijk} v_{i,k} + \frac{1}{2} \epsilon_{ijk} v_{i,k} \quad (\epsilon_{ijk} = -\epsilon_{kji})$$

$$= \epsilon_{ijk} v_{i,k}.$$

Hence,

$$\nabla \times \vec{v} = \vec{w}$$

$$= w_j \vec{e}_j$$

$$= \epsilon_{ijk} v_{i,k} \vec{e}_j. \quad //$$

Laplacian Operator

Definition (2.14): To any scalar field $\phi: \mathbb{E}^3 \rightarrow \mathbb{R}$ we associate another scalar field $\Delta\phi: \mathbb{E}^3 \rightarrow \mathbb{R}$ defined by

$$\Delta\phi := \nabla \cdot (\nabla\phi).$$

To any vector field $\vec{v}: \mathbb{E}^3 \rightarrow V$ we associate another vector field $\Delta\vec{v}: \mathbb{E}^3 \rightarrow V$ defined by

$$\Delta\vec{v} = \nabla \cdot (\nabla\vec{v}).$$

We call $\Delta\phi$ and $\Delta\vec{v}$ the Laplacians of ϕ and $\vec{v}$, respectively.

Result (2.15): Let $\mathcal{B} = \{\vec{e}_i\}$ be an arbitrary frame and let $\vec{v}(\vec{x}) = v_i(\vec{x})\vec{e}_i$. Then,

$$\Delta\phi(\vec{x}) = \phi_{,ii}(\vec{x})$$

and

$$\Delta\vec{v}(\vec{x}) = v_{i, jj}(\vec{x})\vec{e}_i,$$

where (x_1, x_2, x_3) are the coordinates of $\vec{x}$ in $\{\vec{e}_i\}$.

Proof: By Result 2.4 we have $\nabla\phi = \phi_{,i}\vec{e}_i$.
Using Result 2.8

$$\Delta\phi = \nabla \cdot (\nabla\phi) = (\phi_{,i})_{,i} = \phi_{,ii}.$$

By Result 2.6, $\nabla\vec{v} = v_{i,j}\vec{e}_i \otimes \vec{e}_j$. Using Result 2.10)

$$\Delta\vec{v} = \nabla \cdot (\nabla\vec{v}) = (v_{i,j})_{,j}\vec{e}_i = v_{i, jj}\vec{e}_i. //$$

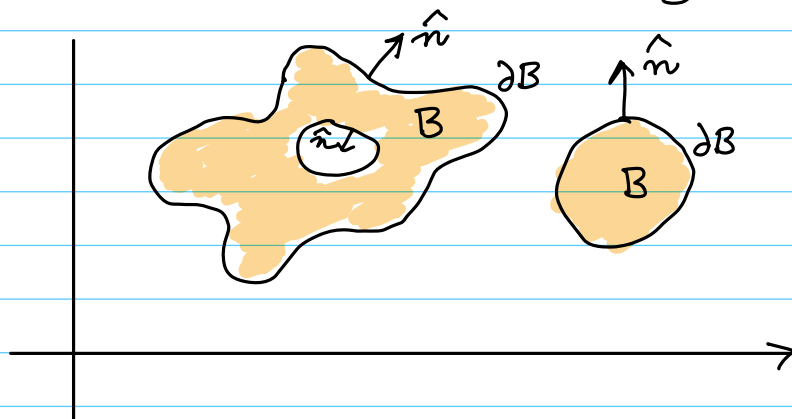
Divergence Theorem

Definition: A region $B \subseteq \mathbb{E}^3$ is called regular iff

i) B consists of a finite number of open, disjoint and bounded components

ii) The bounding surface ∂B is piecewise smooth and consists of a finite number of disjoint components

iii) each component of ∂B is orientable in the sense that it clearly has two sides



B can have holes and disconnected components.

Result (2.16) (Divergence Theorem): let B denote a regular region in $\mathbb{E}^3$ with a piecewise smooth boundary ∂B and consider an arbitrary vector field $\vec{v}: B \rightarrow V$. Then

$$\int_{\partial B} \vec{v} \cdot \hat{n} \, dA = \int_B \nabla \cdot \vec{v} \, dV$$

or, in components

$$\int_{\partial B} v_i n_i dA = \int_B v_{i,i} dV,$$

where $\hat{n}$ is the outward-pointing unit normal field on ∂B .

$\int_{\partial B} v_i n_i dA$ is called the flux of $\vec{v}$ across the surface ∂B .

Result (2.17) let B denote a regular region in $\mathbb{R}^3$ with piecewise smooth boundary ∂B , and consider an arbitrary second-order tensor field $S: B \rightarrow \mathcal{V}^2$. Then

$$\int_{\partial B} S \hat{n} dA = \int_B \nabla \cdot S dV$$

or, in component form,

$$\int_{\partial B} S_{ij} n_j dA = \int_B S_{ij,i} dV,$$

where n is the outward unit normal field on ∂B .

Proof: let $\vec{a} \in \mathcal{V}$ be any constant vector. Then

$$\vec{a} \cdot \int_{\partial B} S \hat{n} dA = \int_{\partial B} \vec{a} \cdot S \hat{n} dA = \int_{\partial B} (S^T \vec{a}) \cdot \hat{n} dA$$

Since $S^T \vec{a}$ is a vector field and

$$\begin{aligned} \nabla \cdot (S^T \vec{a}) &= \frac{\partial}{\partial x_i} (S_{ji} a_j) \\ &= \frac{\partial S_{ji}}{\partial x_i} a_j \end{aligned}$$

$$= (\nabla \cdot \mathbf{S}) \cdot \vec{a},$$

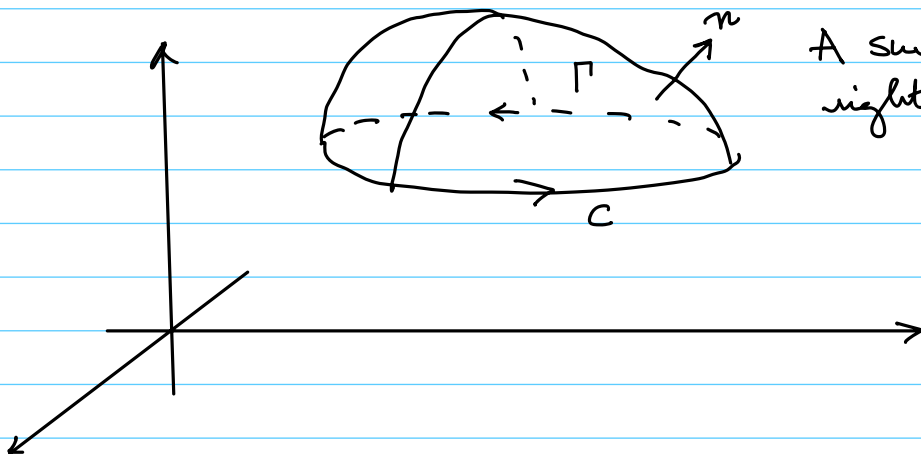
applying the divergence theorem,

$$\begin{aligned} \vec{a} \cdot \int_{\partial B} \mathbf{S} \hat{n} \, dA &= \int_{\partial B} (\mathbf{S}^T \vec{a}) \cdot \hat{n} \, dA \\ &= \int_B \nabla \cdot (\mathbf{S}^T \vec{a}) \, dV \\ &= \int_B (\nabla \cdot \mathbf{S}) \cdot \vec{a} \, dV \\ &= \vec{a} \cdot \int_B \nabla \cdot \mathbf{S} \, dV. \end{aligned}$$

Since $\vec{a} \in V$ was arbitrary,

$$\int_{\partial B} \mathbf{S} \hat{n} \, dA = \int_B \nabla \cdot \mathbf{S} \, dV. \quad ///$$

Stokes Theorem



A surface Γ with
right handed orientation.

Result (2.18) (Stokes' Theorem): let Γ be a surface in $\mathbb{E}^3$ with a piecewise smooth boundary curve $C = \partial\Gamma$ and consider an arbitrary vector field $\vec{v}: \mathbb{E}^3 \rightarrow \mathcal{V}$. Then

$$\int_{\Gamma} (\nabla \times \vec{v}) \cdot \hat{n} \, dA = \int_C \vec{v} \cdot \vec{t} \, ds$$

where $\hat{n}$ is a unit normal field on Γ and $\vec{t}$ is a unit tangent field on C , oriented in the right-handed fashion.

$\int_C \vec{v} \cdot \vec{t} \, ds$ is called the circulation of $\vec{v}$ around the curve C .

Proof: M Spivak, Calculus on Manifolds, Addison-Wesley, 1965. //

The Localization Theorem

Result (2.19): Consider an open set B in $\mathbb{E}^3$ and a continuous function $\phi: B \rightarrow \mathbb{R}$. let Ω denote an arbitrary open subset of B . If

$$(2.1) \quad \int_{\Omega} \phi(\vec{x}) \, dV = 0 \quad \forall \Omega \subseteq B,$$

then

$$\phi(\vec{x}) = 0, \quad \forall \vec{x} \in B.$$

(2.2)

Proof: Assume that (2.2) is true, but (2.1) is false. There is some point $\vec{y} \in B$ for

which $\phi(\vec{y}) \neq 0$. Suppose that $\phi(y) = 2\delta$, where $\delta > 0$. The case $\delta < 0$ can be handled similarly.

Since ϕ is continuous, there is an open neighborhood of $\vec{y}$, call it Ω , such that

$$i) \quad y \ni \Omega \subseteq B$$

$$ii) \quad \phi(\vec{x}) > \delta, \quad \forall x \in \Omega.$$

For this set

$$\int_{\Omega} \phi(x) dV > \delta \text{Vol}(\Omega) > 0,$$

which contradicts (2.1). Hence (2.2) is true. $\parallel$